SECTION 3

Electrolysis

USING ELECTRICAL ENERGY TO DRIVE CHEMICAL REACTIONS

Standards

Electrochemistry: students explain how redox reactions convert chemical energy to electrical energy and back, and apply this to cells, batteries, corrosion, and electrolysis.

Objectives

- How can a spontaneous redox reaction be reversed in an electrochemical cell?
- How do the reactions in the electrolysis of molten sodium chloride compare with those of brine?
- Why is electrolysis important in purifying and producing metals?

Expectations

- Describe how an electrolytic cell differs from a voltaic cell.
- Write the electrode reactions for the electrolysis of molten NaCl and of brine.
- Explain how aluminum is produced and how objects are electroplated.

Section 3

Electrolysis

MAIN IDEA

In electrolysis, a power source causes nonspontaneous reactions to occur in electrochemical cells.

KWS

WHAT I KNOW	WHAT I WANT TO KNOW	SOURCE
		Tyler DeWitt: Electrolysis Tyler DeWitt: Electroplating

Vocabulary

Review

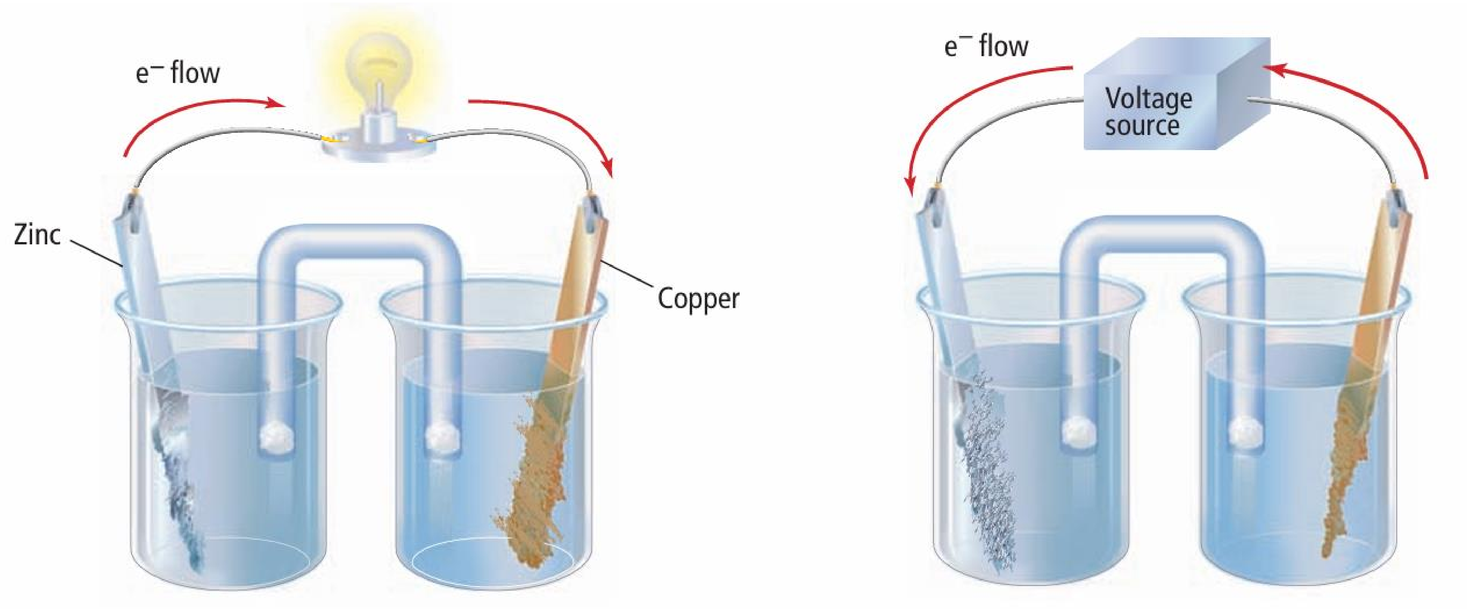
- **redox reaction** — a reaction in which electrons are transferred from one species to another
- **voltaic cell** — converts chemical energy to electrical energy by a spontaneous redox reaction

New

- **electrolysis** — the use of electrical energy to bring about a chemical reaction
- **electrolytic cell** — an electrochemical cell in which electrolysis occurs

Reversing Redox Reactions

- The use of electrical energy to bring about a chemical reaction is called electrolysis.
- An electrochemical cell in which electrolysis occurs is called an electrolytic cell.



WATCH Tyler DeWitt: Electrolysis

Voltaic Cell vs. Electrolytic Cell

	Voltaic cell	Electrolytic cell
Energy change	chemical → electrical	electrical → chemical
Redox reaction	spontaneous	nonspontaneous — must be driven
In the circuit	a load (bulb, motor)	a voltage source
Oxidation occurs at	the anode	the anode
Reduction occurs at	the cathode	the cathode
Zinc–copper example	zinc dissolves, copper deposits	copper dissolves, zinc deposits

Recharging a secondary battery turns it into an electrolytic cell.

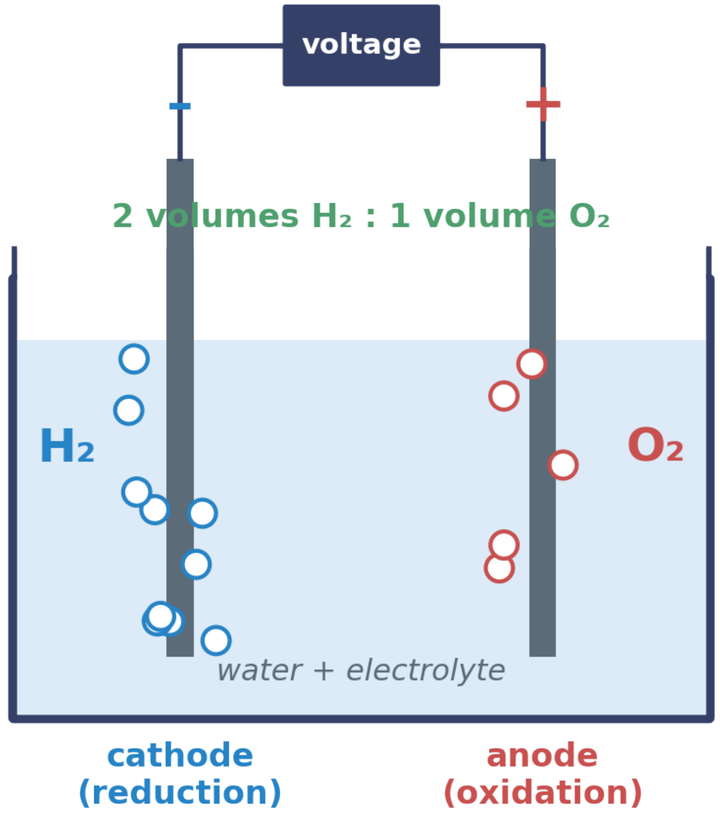
Application: Electrolysis of Water

- Electrolysis of water is one method of obtaining hydrogen gas for commercial use.
- Hydrogen gas forms at the cathode and oxygen gas forms at the anode.
- Twice as much hydrogen as oxygen is produced.

Overall reaction

ELECTRICAL ENERGY IN

$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$



Electrolysis of water

Application: Molten Sodium Chloride

- Electrolysis can separate molten sodium chloride into sodium metal and chlorine gas in a chamber called a Down's cell.
- At the anode, chloride ions are oxidized to chlorine gas.
- At the cathode, sodium ions are reduced to sodium metal.

Down's cell

ANODE

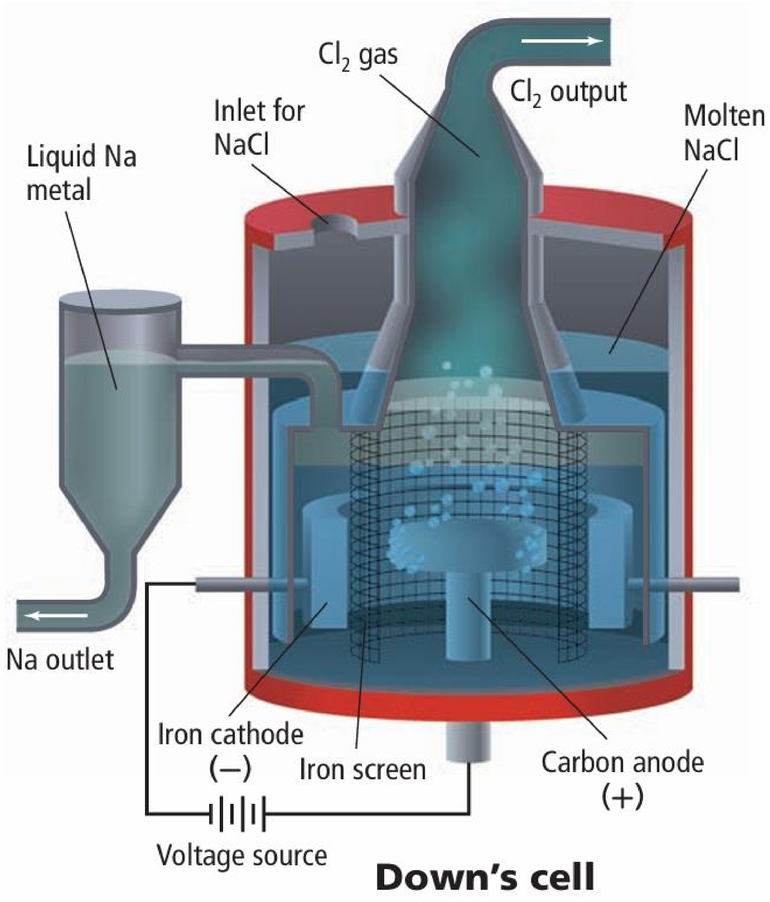
$$2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$$

CATHODE

$$Na^+(l) + e^- \rightarrow Na(l)$$

NET CELL REACTION

$$2Na^+(l) + 2Cl^-(l) \rightarrow 2Na(l) + Cl_2(g)$$



Down's cell

Application: Electrolysis of Brine

In the decomposition of brine, an aqueous solution of NaCl, electrolysis is used to produce hydrogen gas, chlorine gas, and sodium hydroxide.

Possible at the cathode

NOT FAVOURED

$$Na^+(aq) + e^- \rightarrow Na(s)$$

OCCURS

$$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$$

Possible at the anode

OCCURS

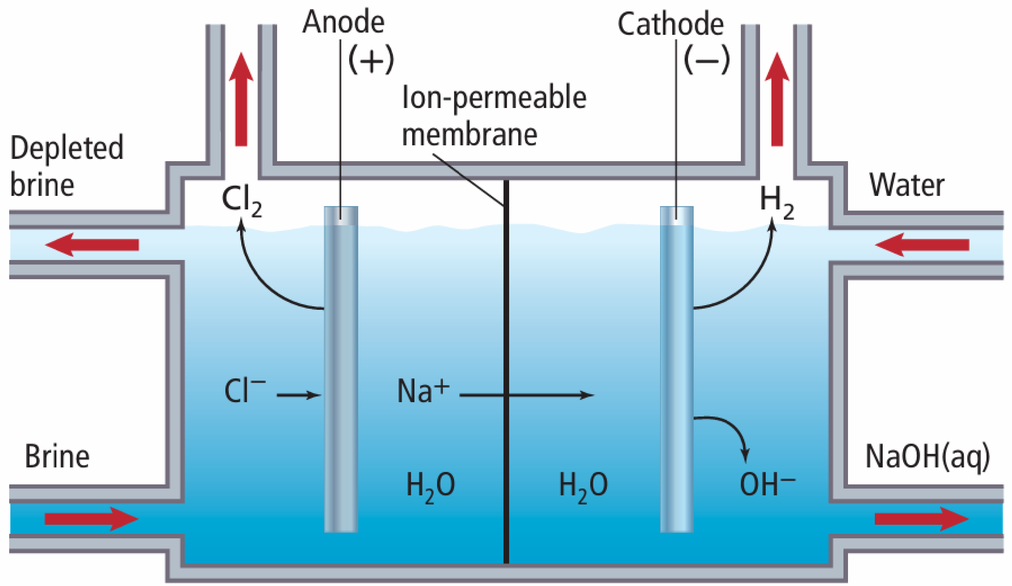
$$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$$

NOT FAVOURED

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$$

OVERALL

$$2H_2O(l) + 2NaCl(aq) \rightarrow H_2(g) + Cl_2(g) + 2NaOH(aq)$$



Electrolysis of brine

Application: Producing Aluminum

- Aluminum is produced by the Hall–Héroult process: aluminum oxide is dissolved in molten cryolite (Na_3AlF_6) and electrolyzed.
- Molten aluminum collects at the carbon-lined cathode; the carbon anodes are slowly used up as they form CO_2.

Hall–Héroult process

CATHODE

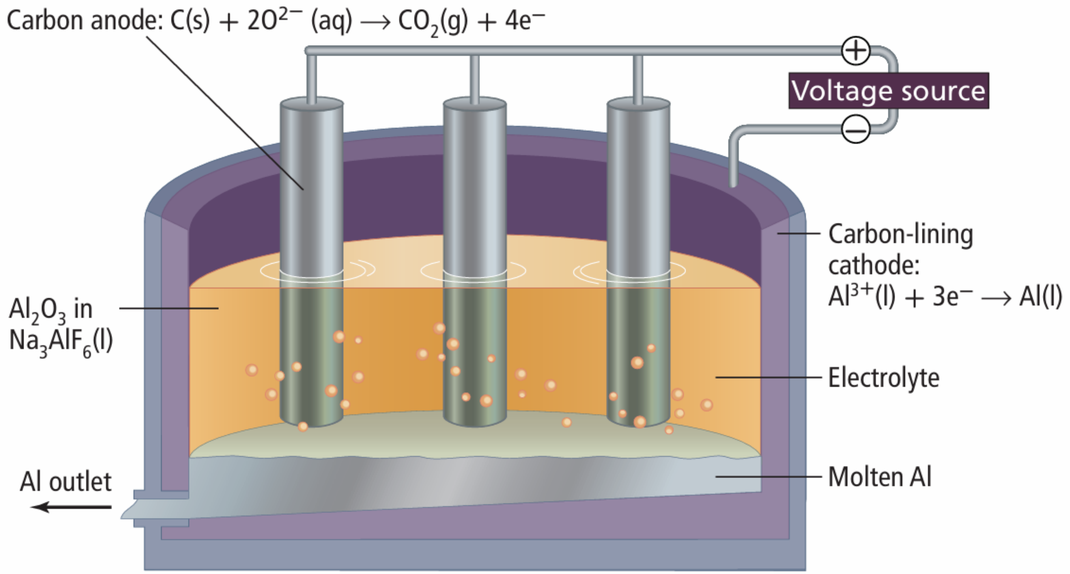
$$Al^{3+}(l) + 3e^- \rightarrow Al(l)$$

ANODE

$$2O^{2-} \rightarrow O_2(g) + 4e^-$$

THEN

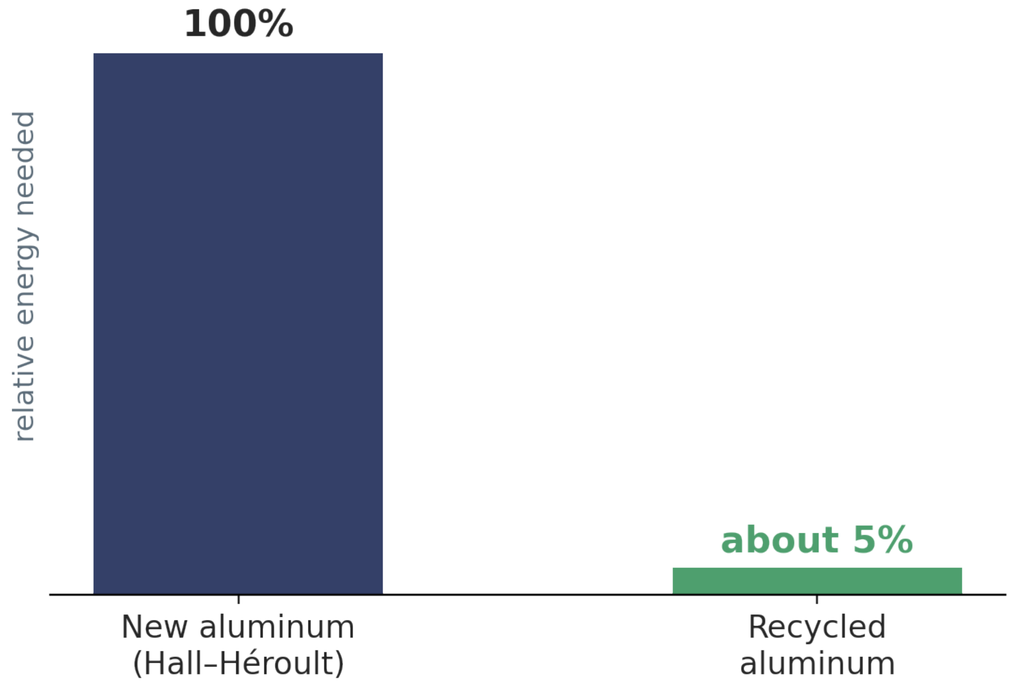
$$C(s) + O_2(g) \rightarrow CO_2(g)$$



Hall–Héroult cell

Why Recycle Aluminum?

- The Hall–Héroult process requires large amounts of energy, which is the primary reason for recycling aluminum.
- Recycled aluminum only has to be melted — it has already been through electrolysis.
- **Extension:** recycling uses only about 5% of the energy needed to make new aluminum from ore.



Approximate energy comparison

Application: Electroplating

- Objects can be electroplated with a metal such as silver.
- The cathode is the object to be electroplated, where reduction occurs.
- The anode is a bar of silver, where silver is oxidized and silver ions are transferred to the cathode.

Silver plating

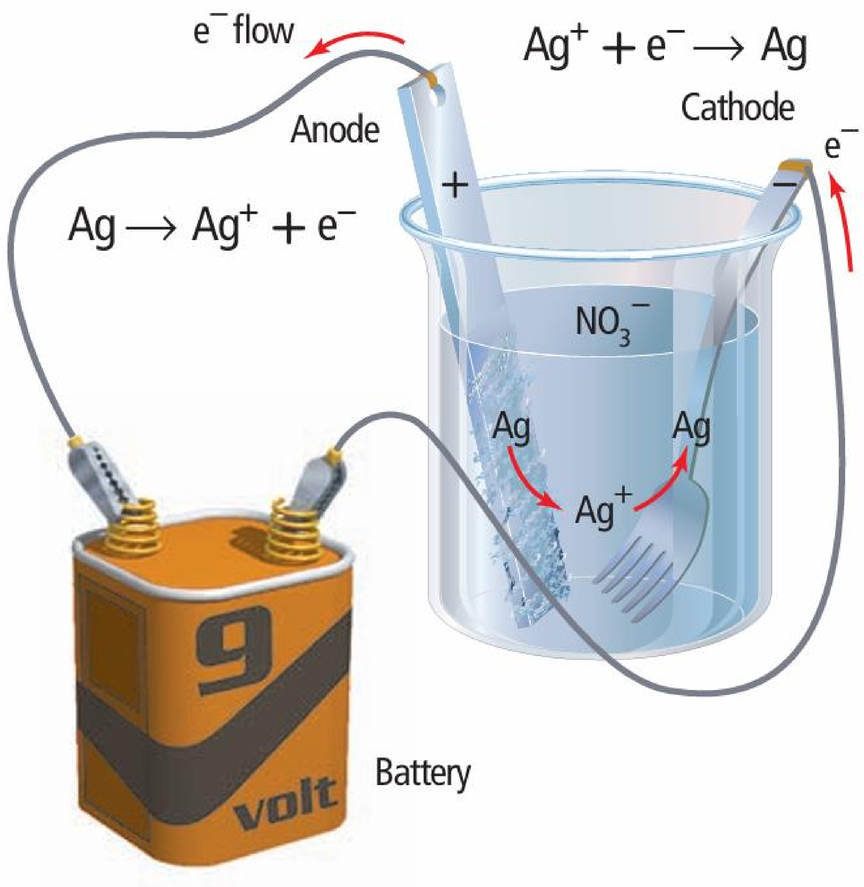
ANODE (SILVER BAR)

$Ag(s) \rightarrow Ag^+(aq) + e^-$

CATHODE (OBJECT)

$Ag^+(aq) + e^- \rightarrow Ag(s)$

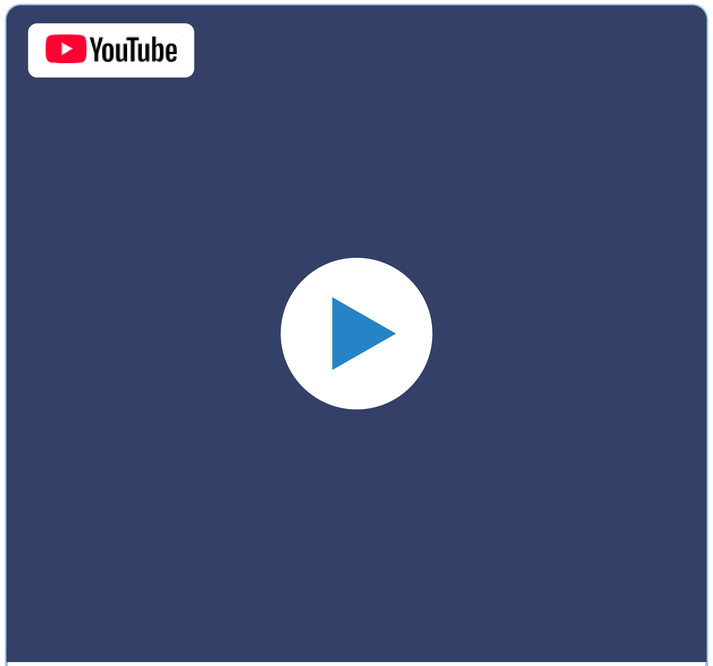
WATCH Tyler DeWitt: Electroplating



Electroplating a fork with silver

Watch & Explore: Electrolysis

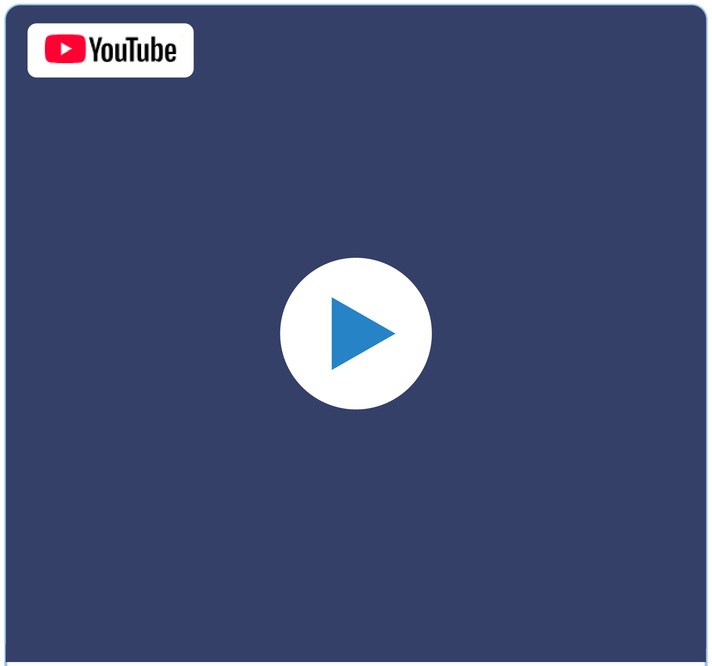
Click a card to open it.



Electrolysis

Tyler DeWitt • 33 min

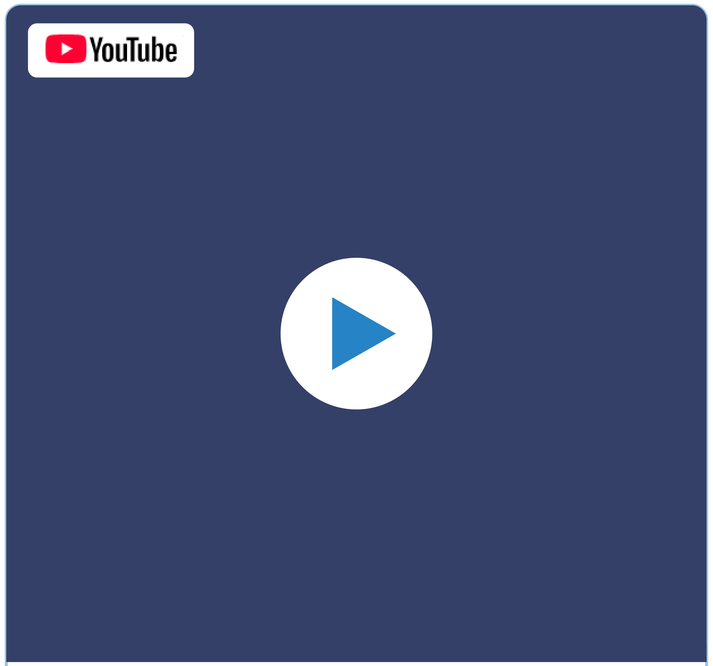
Electrolytic cells with molten NaCl and water as examples.



Electroplating

Tyler DeWitt • 15 min

How a thin layer of metal is plated onto an object.



Introduction to Electrochemistry

Tyler DeWitt • 17 min

Review: how voltaic and electrolytic cells are related.

Checkpoint: Electrolysis

1. Define electrolysis and electrolytic cell.

2. What must be added to a voltaic cell's circuit to reverse its reaction?

3. Name the products of the electrolysis of molten sodium chloride.

4. Which three products are made by the electrolysis of brine?

5. In silver plating, is the object the anode or the cathode?

Quick check — discuss with a partner, then confirm with your teacher.

Lesson Summary

Key Takeaways

- In an electrolytic cell, an outside source of power causes a nonspontaneous redox reaction to occur.
- The electrolysis of molten sodium chloride yields sodium metal and chlorine gas. The electrolysis of brine yields hydrogen gas, sodium hydroxide, and chlorine gas.
- Electrolysis of water is one way to obtain hydrogen gas.
- Electrolysis is used to electroplate objects and to produce pure aluminum from its ore.

Vocabulary & Key Reactions

- **electrolysis** — electrical energy brings about a chemical reaction
- **electrolytic cell** — a cell in which electrolysis occurs
- **Molten NaCl:** $2Na^+ + 2Cl^- \rightarrow 2Na + Cl_2$
- **Brine:** $2H_2O + 2NaCl \rightarrow H_2 + Cl_2 + 2NaOH$
- **Aluminum:** $Al^{3+} + 3e^- \rightarrow Al$
- **Plating:** $Ag^+ + e^- \rightarrow Ag$ at the cathode

Evaluation: Electrolysis

Complete independently. Review the summary if you get stuck, then check your answers with your teacher.

1. Define electrolysis and relate the definition to the spontaneity of redox reactions.

2. Explain why the products of the electrolysis of brine and the electrolysis of molten sodium chloride are different.

3. Describe how a voltaic cell can be changed into an electrolytic cell.

4. Explain why recycling aluminum saves so much energy.

5. In an electroplating cell, identify the anode and the cathode and write the half-reaction at each.

6. **Challenge:** You want to plate an iron key with copper. Sketch the electrolytic cell, label the anode, cathode, electrolyte, and direction of electron flow, and write both half-reactions.

Self-Evaluation Rubric

4 – Confident mastery • 3 – Solid grasp, minor gaps • 2 – Needs review • 1 – Revisit the lesson